Natural Language - Project report

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1 Introduction

We assign clinical records to one of 12 medical specialties, with 2,662 training and 398 test records adapted from a real Kaggle dataset[1]. Each record has a *description*, a *sample name*, *keywords* and the *transcription* of the clinical note. The transcription holds most of the text (median 387 words, against 4–19 for the other fields) but decides the least: identical transcriptions carry different specialties, while the first keyword names the specialty and the section headers (of transcription) reveal the type of document (Section 2.1). Hence our research question (RQ):

RQ: *Does a short, information-dense input (keywords, sample name, section headers) classify clinical records better than the complete record, and does it still do so when the keywords are missing?*

Short inputs avoid dilution: in a term frequency–inverse document frequency (TF-IDF) vector, normalised to unit length, the first keyword weighs little among the hundreds of words of a complete record. They also carry a risk: a model that relies on the keywords learns a shortcut (Geirhos et al., 2020), copying the first keyword, which fails on the 19.6% of test records that have no keywords. We therefore compare short and complete inputs with classifiers from Multinomial Naive Bayes (MNB) to a linear Support Vector Machine (SVM), and evaluate them separately on records with and without keywords.

[1] `https://www.kaggle.com`

2 Data

2.1 Data Analysis

The classes of the 2,662 training records are imbalanced (Table A1): Surgery has 890 records (33.4%) and Dermatology 25 (0.9%). Always predicting Surgery reaches 33.4% accuracy but F1 = 0 for the other 11 classes, failing the shared-task requirement of an F1 of at least 0.25 for every class.

O1: one unclosed quote, two kinds of damage. The 13 lines of the training file without the four ";" of a record come from seven truncated lines that hold only a label, an opening quote and the start of a description (the other six are where those quotes close). A CSV reader takes everything up to the closing quote as one field. In four cases, this merges up to 12 records into one description; in the other three, the merged text is cut at exactly 32,759 characters (just under Excel's cell limit) and the rest continues on a new line, read as a record whose label is text ("d is open about. . . "). Hence `pandas` returns 2,616 rows and 15 "labels". Parsing the file line by line, we drop the seven fragments (their text occurs nowhere else) and rejoin the cut text with the last line, recovering 46 records, having in total 2,662 records.

O2: the first keyword is the label in all 2,219 training records with keywords ("*neurosurgery*,

herniated nucleus pulposus, . . . "), and $\approx 80\%$ of the test records start with a specialty name; the others have no keywords. Some classes lack keywords far more often than others (Table A1): 66% of Psychiatry-Psychology records against 4% of Ophthalmology.

O3: the same transcription under several labels. 58% of the training records have the same transcription as another record. Together they hold 733 shared transcriptions, and 732 of them appear under different labels; the copies differ only in the label and the first keyword. For example, one "Anterior Cervical Discectomy" note appears three times, as Neurosurgery, Orthopedic and Surgery. The reason is that the labels follow two criteria: some name the type of document (84% of Surgery records are procedure notes, with headers such as "ANESTHESIA:"; 80% of Radiology records are imaging reports, with "IMPRESSION:"), others a body system (Orthopedic, Ophthalmology), so one operation report fits both.

We also found minor noise: in 150 records the keywords end with a website disclaimer ("NOTE: These transcribed medical transcription sample reports. . . " inside keywords), 24 records are templates with an empty transcription ("EGD Template - 1"), and names are replaced by placeholders ("I spoke with Dr. X").

2.2 Data Augmentation

After an 80/20 split, the training part has 712 Surgery records but only 20 Dermatology records. We balance it by adding new versions of random records of each other class until every class has 712 records; validation and test records are never changed.

The new versions are made by *clinical EDA*, our adaptation of Easy Data Augmentation (EDA; Wei and Zou, 2019). EDA changes a text with four random word operations; we keep two and adapt them to clinical notes. (i) *Sentence dropout*: instead of single words, we delete whole sentences, each with probability $p = 0.2$, so the remaining sentences keep their meaning (*no fracture* never loses its *no*). (ii) *Abbreviation replacement*: instead of WordNet synonyms, which cover little medical vocabulary, we swap 30 frequent abbreviations with their full form, or the reverse, each occurrence with $p = 0.5$ (MRI $\leftrightarrow$ *magnetic resonance imaging*). We drop random swap and insertion, as they create word sequences that never occur in real notes.

On the complete record, balancing raises the macro-F1 from 0.16 to 0.52 for MNB and from 0.28 to 0.57 for Logistic Regression (LR) (Table A2). How the new rows are written matters little: exact copies and the Synthetic Minority Over-sampling Technique (SMOTE; Chawla et al., 2002) score like clinical EDA, because a new row stays 94% similar to its original (cosine similarity of their TF-IDF vectors). We therefore use clinical EDA for MNB. For the linear models we use class weights instead, which balance the classes without new rows: the error on each record is multiplied by $n/(12\, n_c)$, where n_c is the size of its class and n that of the training part (LR: 0.570, the best score in Table A2).

3 Models

4 Experimental Setup

4.1 Parameters/Hyperparameters

5 Results

6 Discussion

7 Future Work

Bibliography

Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357.

Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., and Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673.

Wei, J. and Zou, K. (2019). EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6382–6388.

Appendix A: Extra Figures and Tables

Table A1: Training set per class: records (n), % without keywords, % whose transcription is shared with another record, median words per transcription and main document type (P: procedure, I: imaging or test report, C: consultation) with its %.

Class	n	No kw	Shared	Words	Type
Surgery	890	7	61	405	P 84
Cardio-Pulm	308	25	58	358	P 37
Orthopedic	300	16	71	505	P 60
Radiology	222	9	73	230	I 80
Gen. Medicine	211	42	8	420	C 74
Gastro	195	15	60	285	P 54
Neurology	184	27	55	419	I 48
Obs-Gyn	135	17	65	387	P 52
Neurosurgery	78	15	92	538	P 87
Ophthalmology	73	4	58	323	P 68
Psych	41	66	20	661	C 66
Dermatology	25	20	48	254	P 44
All	2,662	17	58	387	P 52

Table A2: Balancing strategies on the complete record (TF-IDF of word 1–2-grams): accuracy (Acc), macro-F1 (F1) and lowest per-class F1 (Min) on validation, mean of 3 stratified 80/20 splits (standard deviation ≤ 0.031 for Acc and F1). Undersampling keeps 20 records per class; original EDA deletes and swaps 10% of the words; SMOTE interpolates TF-IDF vectors. Best value per column in bold.

	MNB			LR		
Balancing	Acc	F1	Min	Acc	F1	Min
None	.442	.161	.000	.462	.280	.000
Undersampling	.424	.436	.257	.475	.490	.319
Class weights	.458	.466	.271	**.532**	**.570**	**.324**
SMOTE	.473	.491	.271	.503	.534	.201
Exact copies	.487	.509	.257	.523	.546	.243
Original EDA	.488	.513	**.278**	.502	.524	.188
Clinical EDA	**.490**	**.516**	.274	.513	.533	.242

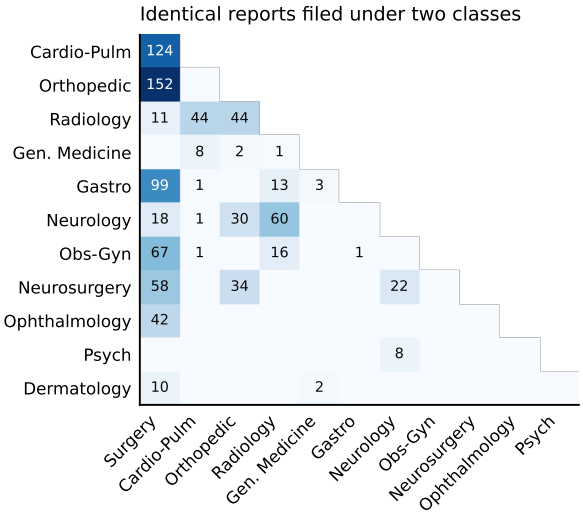


Figure A1: Identical transcriptions with different labels, per pair of classes (training set, O3): each cell counts the transcriptions that appear, word for word, under both classes; e.g., 152 transcriptions appear both as Surgery and as Orthopedic.

A Evaluation and tips (remove this section before submission)

Scores by section:

- (1.5) Introduction: beyond the classification task itself, **explicitly** identify the research question your group wants to answer. For instance: how does classical machine learning compare to current language models on clinical text? Which data augmentation strategy is most effective for highly imbalanced medical datasets? This research question should guide your model choices, your experimental setup, and your discussion. It defines the **story of the paper**. The remaining sections of the paper should be read as an attempt to answer it.
- (1.5) Data Analysis: describe the dataset (class distribution and relevant statistics); provide at least 2 concrete observations about patterns or anomalies you found by looking at the data. Generic descriptions score zero.
- (2.0) Data Augmentation: you should do some kind of data augmentation. Clearly describe your strategy.
- (4.0) Models: clearly describe the models you built and the choices you made, considering the research question you have identified previously (you should describe at least 3 models) (1.0). When describing your models, keep replicability in mind (1.0): would someone be able to reproduce your results just by reading this section? We will also reward creativity (2.0): did you explore different approaches and make thoughtful choices, or did you simply run code you found online?
- (1.0) Experimental Setup: provide clear information regarding the used datasets, splits (which result in your own test set(s)), evaluation metrics used and hyperparameters (if applicable).
- (1.5) Results: provide the results of your best models, and also results by label of your best model. You should also present a comprehensive confusion matrix of one of your models (not necessarily the best one). The given test set (test_no_labels.txt) should ****only**** be used to generate the file results.txt.
- (3.0) Discussion: show us that you have properly analysed the dataset and the obtained outputs (not just by looking at statistics or confusion matrices). Try to explain the most common errors. Present and discuss at least 3 concrete examples of misclassified instances from your own test set (include the text excerpt, the predicted label, the correct label, and your interpretation). Generic statements score zero. Also, explain how results align with the research question you have identified in the introduction.
- (0.5) Future Work: if more time was given, explain what you would do to improve your system, considering the research question identified in the introduction.
- (1.0) References: Identify all papers, external sources, and pre-trained models used; also explicitly declare how have you used LLMs.

Tips:

- Label figures and tables.
- Use the correct quotation marks in Latex (“bla-bla”).
- If you use a figure or table, refer to it.
- If you say things such as “The dataset is unbalanced”, explain why (facts).
- Use formal English (so, no “it’s”, “they’re”, etc.).

- If you mention a hyperparameter, explain it (briefly).
- If you use an acronym, use it consistently (check how to do it in latex).
- Use labels that are easy to be read in the Confusion Matrix.
- Cite your sources (papers) or add footnotes with URLs for computational resources.
- Do not use subjective adjectives (e.g.: amazing work, nice concept, ...). This is science.